

Study guide

- Know Theorem 6.1 about (i) how to compute orders of elements in $G \times H$ and (ii) how to tell when $G \times H$ is cyclic.
- Understand the proof of Theorem 6.1, which we've seen a few variations of recently.
- Definition: even and odd permutations.
- How do you tell even from odd permutations in S_4 using a tetrahedron?
- How do you factor an r -cycle into transpositions? How many transpositions are needed?
- How do you determine the sign (odd or even) of a permutation?
- Definition: the *alternating group* A_n . Why is it a subgroup of S_n ?
- Know the *statement*: every permutation is odd or even, *but not both*. (Theorem 8.4). This will be proved next week.

1. Find all subgroups of the group $\mathbb{Z}_2 \times \mathbb{Z}_4$.
2. Determine the largest possible order of an element of S_9 . (A problem from PSet 4 may be useful.)
3. Does A_6 have an element of order 6? Does A_7 ? If so, give an example. If not, prove that it is impossible.
4. Suppose that H is a subgroup of S_n . Prove that either all elements of H are even permutations, or exactly half of the elements of H are even permutations.

Hint You can solve this however you like, but here is one strategy you might try. Let $H_0 \subseteq H$ be the set of even permutations in H and let H_1 be the set of odd permutations in H . If H_1 is empty then we are done, so suppose it has some element $g \in H_1$. Now consider the function $f : H_0 \rightarrow H_1$ given by $f(x) = gx$. Show that this function is a bijection. You might also read the proof on p. 72 of the textbook that A_n has exactly half of the elements of S_n , which is a special case of this problem.

5. Suppose that G is a group of order n , and k is an integer such that $(k, n) = 1$.
 - (a) Prove that there exists an integer ℓ such that for all $g \in G$, $(g^k)^\ell = g$.
 - (b) Deduce from (a) that the function $f : G \rightarrow G$ defined by $f(g) = g^k$ is bijective, and give an explicit description of its inverse function.
 - (c) Deduce that if G is a finite group of odd order, then every element of G has a unique “square root” (i.e. for every $g \in G$, there exists a unique $h \in G$ such that $h^2 = g$).

Note: One way to phrase this fact is that it is possible to compute unique “ k th roots” in a group G , assuming that k is relatively prime to $|G|$. This fact has an important application in the design of the commonly-used RSA cryptosystem.